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Communication

Nickel-Catalyzed Enantioselective Arylative Activation of Aromatic C−O Bond

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Cite This: J. Am. Chem. Soc. 2021, 143, 18380−18387
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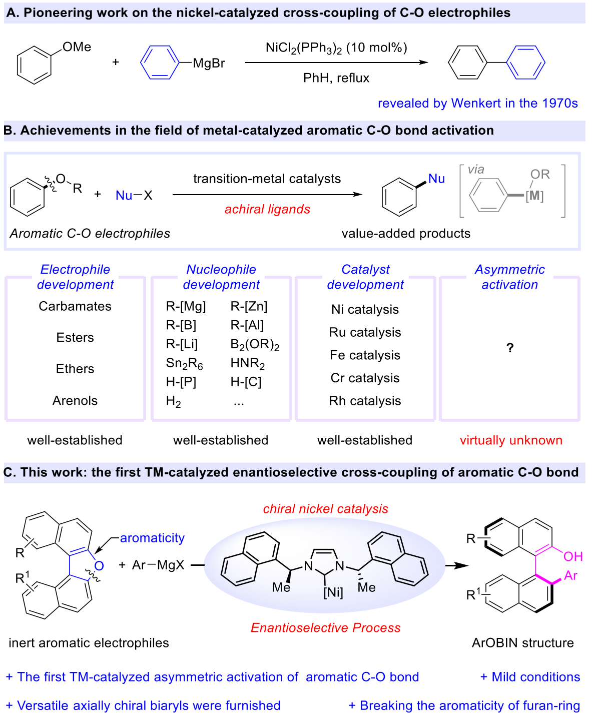
ABSTRACT: The pioneering nickel-catalyzed cross-coupling of C−O electrophiles was unlocked by Wenkert in the 1970s; however, the transition-metal-catalyzed asymmetric activation of aromatic C−O bonds has never been reported. Herein the first enantioselective activation of an aromatic C−O bond is demonstrated via the catalytic arylative ring-opening cross-coupling of diarylfurans. This transformation is facilitated via nickel catalysis in the presence of chiral *N*-heterocyclic carbene ligands, and chiral 2-aryl-2′-hydroxy-1,1′-binaphthyl (ArOBIN) skeletons are delivered axially in high yields with high ee. Moreover, this versatile skeleton can be transformed into various synthetic useful intermediates, chiral catalysts, and ligands by using the CH- and OH-based modifiable sites. This chemistry features mild conditions and good atom economy.

Advancing efficient catalytic processes to forge value-added functional molecules by using easily available and natural abundant feedstock represents an attractive goal in synthetic chemistry.[1] To this end, the aromatic C−O bond, which is widespread in biomass and natural products, was developed as an environmentally benign alternative to conventional organohalides in transition-metal-catalyzed cross-coupling reactions. Pioneered by Wenkert's work[2] (Scheme 1A), extensive efforts were made, and different strategies, including electrophile development, nucleophile development, and catalyst development, were established for the efficient utilization of the aromatic C−O bond in organic synthesis (Scheme 1B).[3] For example, various oxygen-based compounds, such as carbamate,[4] ester,[5] and ether,[6] were demonstrated as suitable electrophiles in the transition-metal-catalyzed cross-coupling reactions. Of particular note, direct cross-couplings of arenol were also unlocked recently.[7] In addition, different types of nucleophiles were also successfully applied to the cross-coupling of C−O electrophiles to construct a carbon−carbon bond[8] and a carbon−heteroatom bond.[9] Moreover, catalysts used in aromatic C−O bond activation have been extended from the original nickel catalyst[10] to ruthenium,[11] iron,[12] rhodium,[13] and chromium catalysts.[14] Indeed, these strategies largely improved the application potential of C−O electrophiles in organic synthesis; however, to the best of our knowledge, the transition-metal-catalyzed asymmetric cross-coupling of aromatic C−O electrophiles is still undeveloped (Scheme 1B).

Scheme 1. Rational Design of the Atroposelective Cross-Coupling of an Inert Aromatic C−O Bond



Triggered by the pivotal role of axially chiral biaryl scaffolds in natural products,[15] pharmaceuticals,[16] and asymmetric synthesis,[17] we aimed to investigate the synthesis of axially chiral biaryl compounds via enantioselective aromatic C−O bond activation. Our strategy toward this goal was based on transition-metal-catalyzed enantioselective ring-opening cross-coupling reactions, which have been established as powerful tools for the preparation of functionalized axially chiral biaryl skeletons.[18] For example, pioneered by Bringmann's work,[19]

Received: September 14, 2021
Published: October 27, 2021

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https://doi.org/10.1021/jacs.1c09797
J. Am. Chem. Soc. 2021, 143, 18380−18387

the transition-metal-catalyzed asymmetric reduction of biaryl lactones for the synthesis of functionalized axially chiral biaryl compounds has been well studied.[20] Hayashi and coworkers demonstrated the asymmetric ring-opening of diarylthiophene using a chiral phinooxazoline ligand.[21] Later on, the same group found that the palladium-catalyzed ring-opening carbonylation of binaphthaleneiodonium salt could occur via an enantioselective process by using (*R*)-BINAP as the ligand, whereas only 28% ee was obtained.[22] Inspired by this work, Gu and coworkers systematically investigated various enantioselective ring-opening cross-couplings of cyclic diaryliodoniums salts.[23] More recently, the enantioselective cross-couplings of 9*H*-fluoren-9-ols and silafluorenes via a ring-opening process were also revealed by Gu and coworkers.[24] Despite these advances, transition-metal-catalyzed enantioselective ring-opening cross-coupling via aromatic C−O bond activation has remained untouched, likely due to the lack of a proper chiral ligand that can control the enantioselectivity during the cleavage of an aromatic C−O bond with high bond dissociate energy.[25] By utilizing a chiral *N*-heterocyclic carbene (NHC) ligand, we herein demonstrate the first enantioselective ring-opening Kumada coupling of a dinaphthofuran derivative for the synthesis of axially chiral biaryl skeletons via C−O bond activation (Scheme 1C). This enantioselective transformation occurs under mild conditions with good atom and step economy and offers a straightforward route to construct versatile axially chiral 2-aryl-2′-hydroxy-1,1′-binaphthyl (ArOBIN) skeletons, which have been found in a myriad of privileged chiral catalysts and ligands (Figure 1).[26]

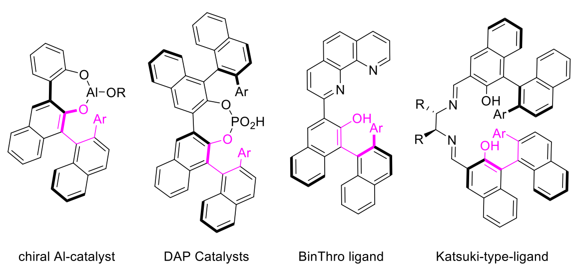


Figure 1. Selected catalysts and ligands derived from ArOBIN.

Finding a unique ligand that can balance the reactivity and the enantiocontrol during the C−O bond cleavage was the key point to realize in this challenging project. Thus by choosing the cross-coupling of dinaphtho[2,1-*b*:1′,2′-d]furan (**1a**) with phenylmagnesium chloride (**2a**) as the model reaction, we first evaluated the reactivity. Despite the fact that the combination of Ni(cod)$_2$ and PCy$_3$ has been proven as one of the most efficient catalytic systems for aromatic C−O bond activation,[3a] it presented low catalytic activity for the cross-coupling of substrate **1a** with the Grignard reagent. The lower reactivity was likely aroused by the aromaticity of the furan ring (see the Supporting Information).[27] This phenomenon was further demonstrated by the evaluation of chiral phosphine ligands (**L1**−**L3**) (Figure 2). The recent development of NHC ligands inspired us to design and screen chiral NHC ligands for the designed cross-coupling (Figure 2).[13,28] Indeed, a nearly quantitative yield and 32% ee were obtained when chiral NHC ligand **L4** was used. This result led us to modify the ligand **L4**. On the basis of the modification of the R′ group, ligand **L5** was prepared by using an ethyl group to replace the methyl group and submitted to the designed enantioselective

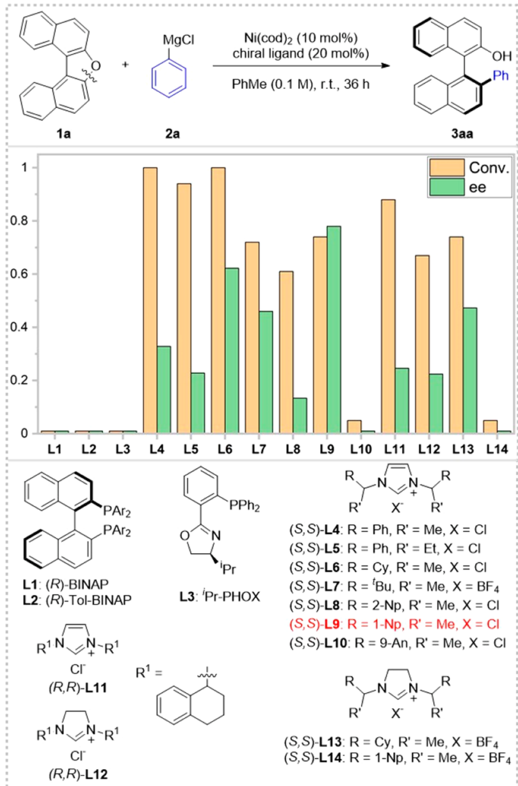


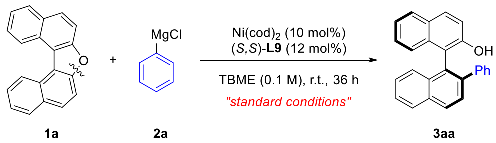
Figure 2. Preliminary evaluation of ligands. The reactions were conducted by using substrate (**1a**) (0.1 mmol), Ph-MgCl (**2a**) (3.0 equiv), Ni(cod)$_2$ (10 mol %), and ligand (20 mol %) in toluene (0.1 M) at room temperature (r.t.) for 36 h. The conversion was determined by ^{1}H NMR using CH_2Br_2 as the internal standard, and the ee was determined by using chiral high-performance liquid chromatography (HPLC).

cross-coupling, where a lower ee was observed. Therefore, we turned to tune the R group and prepared ligands **L6**−**L10**. To our delight, 80% ee and 74% NMR conversion were provided when **L9** was used as the ligand. Further modifications based on ligand **L9** were conducted, and the ligands **L11**−**L14** were synthesized, which failed to improve the enantioselectivity.

By using **L9** as the optimal ligand, we conducted further extensive experiments and revealed that a reaction mixture that contained Ni(cod)$_2$ (10 mol %) and **L9** (12 mol %) in methyl *tert*-butyl ether (MTBE) (0.1 M) at room temperature for 36 h could efficiently facilitate the targeted enantioselective cross-coupling and furnished the desired product **3aa** in 92% isolated yield with 88% ee (Table 1, entry 1). The control experiments demonstrated the pivotal roles of the catalyst and ligand in the nickel-catalyzed enantioselective cross-coupling via inert aromatic C−O bond activation (entries 2 and 3). Interestingly, the air-stable NiCl$_2$(dme), which was proven to be an active catalyst in the reported works on inert C−O activation, provided the desired product **3aa** in lower yield with comparable ee (entry 4). The effect of the solvents was carefully examined, and it was revealed that polar solvents were unsuitable for this transformation (entries 5−8). Additionally, the Grignard reagents that coordinated with different anions were submitted to the enantioselective cross-coupling reaction but proved to be unsuitable coupling partners (entries 9 and 10). Inspired by these results, various salts were used as additives and applied into this transformation but failed to

https://doi.org/10.1021/jacs.1c09797
J. Am. Chem. Soc. 2021, 143, 18380−18387

Table 1. Reaction Conditions Optimization[a]



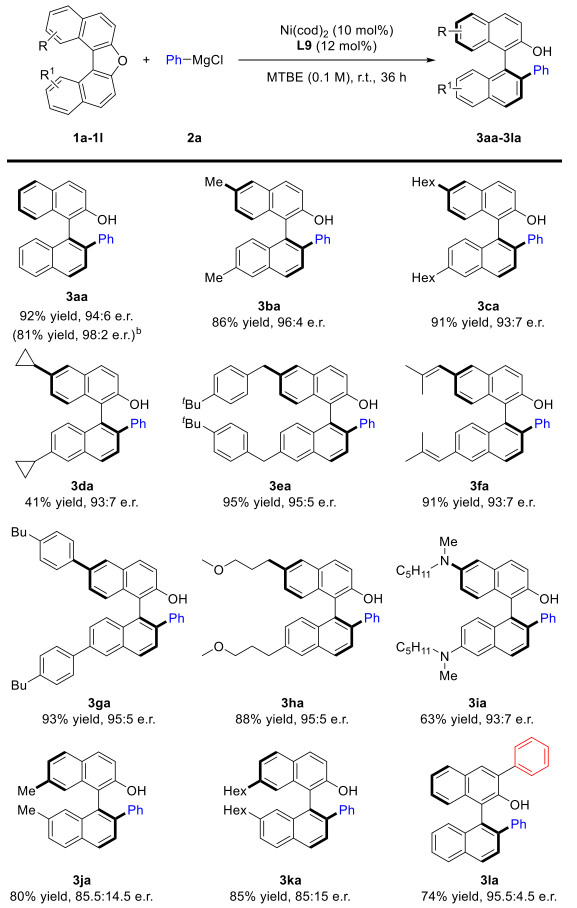
entries	variation from the standard conditions	conv. (%)	e.r.
1	none	91 (92)	94:6
2	$Ni(cod)_2$ was omitted	0	n.d.
3	(*S*,*S*)-**L9** was omitted	58	49:51
4	$NiCl_2$(dme) instead of $Ni(cod)_2$	43	93.5:6.5
5	PhMe instead of TBME	80	90:10
6	2-Me-THF instead of TBME	27	84.5:15.5
7	DME instead of TBME	<10	n.d.
8	CPME instead of MTBE	83	92.5:7.5
9	PhMgBr instead of PhMgCl	96	89.5:10.5
10	PhMgI instead of PhMgCl	100	91:9
11	LiF (2.0 equiv) was added	94	92.5:7.5
12	LiCl (2.0 equiv) was added	93	91:9
13	NaCl (2.0 equiv) was added	95	92:8
14	$MgCl_2$ (2.0 equiv) was added	91	93:7

[a]Reactions were conducted by using C−O electrophiles **1a** (0.1 mmol), Ph-MgCl (**2a**) (3.0 equiv), Ni catalyst (10 mol %), and **L9** (12 mol %) in solvent (0.1 M) at room temperature (r.t.) for 36 h. The conversion was determined by ^{1}H NMR using CH_2Br_2 as the internal standard. Isolated yields are reported in parentheses, and the e.r. (S/R) was determined by using chiral HPLC.

improve the efficiency and enantioselectivity of this transformation (entries 11−14).

With the well-established reaction conditions in hand, we set out to explore the generality of this methodology by applying various C−O electrophiles into the nickel-catalyzed enantioselective cross-coupling reaction (Table 2). The enantioselective ring-opening cross-coupling of C−O electrophiles that bear double substituents at 6- and 6′-postions were primarily conducted (**3ba**−**3ia**). As expected, the alkyl groups, such as methyl and hexyl, did not affect the efficiencies or enantioselectivities of this transformation and offered the axially chiral biaryls **3ba** and **3ca** in high yields with high ee. Surprisingly, compound **3da** was forged in low yield (41%) when a cyclopropyl group was introduced to the C−O electrophile, although a comparable enantiocontrol could be reached. Moreover, the C−O electrophiles that bear benzyl, alkenyl, and phenyl groups were demonstrated as suitable substrates and delivered the desired biaryl atropisomers (**3ea**−**3ga**) in high yields with high ee. It is noteworthy that an sp^3 C−O bond (**3ha**) and an sp^2 C−N bond (**3ia**) could be well tolerated under standard conditions and delivered the functional products with satisfying results. Furthermore, the substitution effect at the 7- and 7′-positions of C−O electrophiles was also investigated, and the corresponding products **3ja** and **3ka** were observed in high yields with moderate ee. In addition to the symmetric double-substituted substrates, a substrate that bears one substituent on one side was prepared and submitted to the nickel-catalyzed enantioselective cross-coupling. To our delight, the desired product (**3la**) was isolated in 74% yield with 91% ee. During this process, only the cleavage of the C−O bond apart from phenyl group was observed, and the C−O bond adjacent to the phenyl group remained untouched. This regioselectivity may be aroused by the steric effect.

Table 2. Scope of C−O Electrophiles for the Atroposelective Cross-Coupling Reactions[a]



[a]Reactions were conducted by using C−O electrophiles **1a**−**1l** (1.0 equiv), Ph-MgCl (**2a**) (3.0 equiv), $Ni(cod)_2$ (10 mol %), and **L9** (12 mol %) in MTBE (0.1 M) at room temperature (r.t.) for 36 h. Isolated yields are reported, and the e.r. (S/R) was determined by using chiral HPLC. [b]Yield and e.r. after recrystallization from hexane/EtOAc.

We further proceeded to investigate the nickel-catalyzed enantioselective ring-opening cross-coupling reactions of 6,6′-dimethyl-dinaphtho[2,1-*b*:1′,2′-d]furan (**1b**) with various Grignard reagents. As shown in Table 3, both Grignard reagents that bear a methyl group at the para and meta positions were demonstrated as suitable nucleophiles and furnished the corresponding products **3bb** and **3bc** in high yields with high ee; however, because of the steric hindrance, poor reactivity was observed with the Grignard reagent bearing a methyl group at ortho position (**3bd**). Additionally, nucleophiles bearing ethyl (**3be**), *tert*-butyl (**3bf**), and benzyl (**3bg**) groups were also submitted to the catalytic system, and the desired products were obtained in good yields with high ee. To our delight, various carbon-heteroatom bonds, including sp^2 C−Si (**3bh**), sp^2 C−N (**3bi**), sp^2 C−O (**3bj**), and homobenzylic C−O (**3bk**) bonds, could be well tolerated and furnished the corresponding products with satisfied results, thus providing the chance for orthogonal functionalization to build complex molecules.[3d,29] In addition to the Grignard reagents that bear a single substituent, multisubstituted nucleophiles were also examined and delivered the targeted axially chiral biaryls in moderate to high yields with high ee

https://doi.org/10.1021/jacs.1c09797
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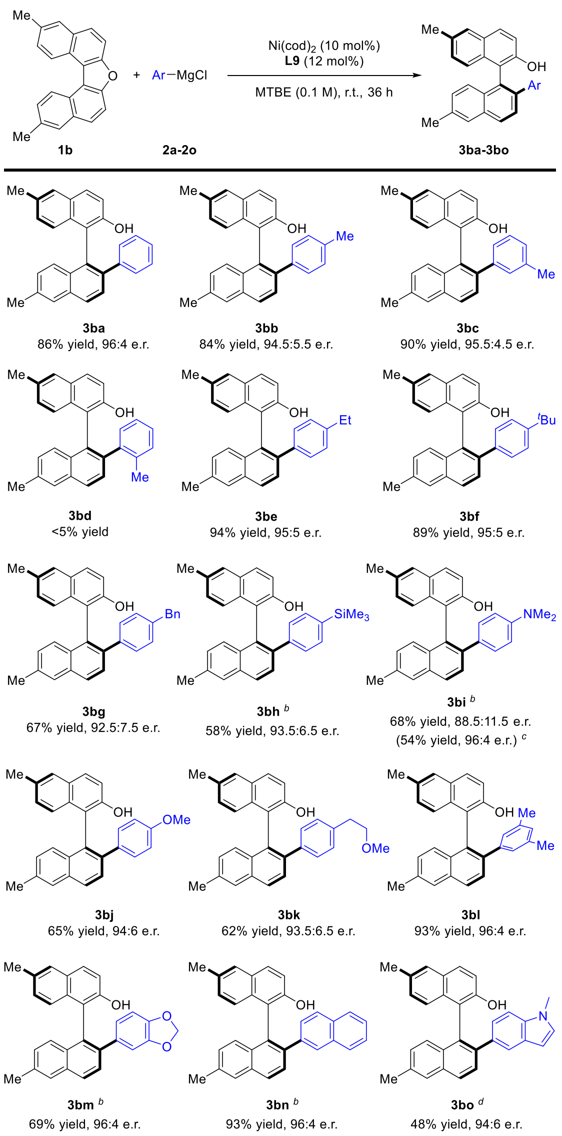
Table 3. Scope of Aryl Grignard Reagents for the Atroposelective Cross-Coupling Reaction[a]

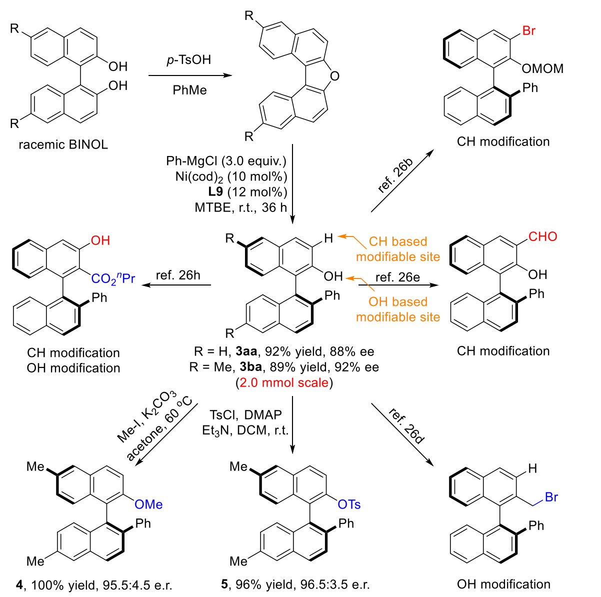


[a]Reactions were conducted by using C−O electrophile **1b** (1.0 equiv), Ar-MgCl (3.0 equiv), Ni(cod)$_2$ (10 mol %), and **L9** (12 mol %) in MTBE (0.1 M) at room temperature (r.t.) for 36 h. Isolated yields are reported, and the e.r. (S/R) was determined by using chiral HPLC. [b]Ar-MgBr (3.0 equiv) was used in the presence of $MgCl_2$ (4.0 equiv). [c]Yield and e.r. after recrystallization from hexane/EtOAc. [d]Ar-MgBr (3.0 equiv) and cyclo-pentyl methyl ether (CPME) was used as the solvent.

(**3bl** and **3bm**). To obtain more axially chiral ArOBIN skeletons, we submitted 2-naphthyl magnesium bromide to the catalytic system, and the product (**3bn**) was obtained in 93% yield with 92% ee. It is worth noting that the enantioselective cross-coupling with the heterocyclic Grignard reagent could furnish the corresponding product **3bo** with 87% ee, whereas only a 48% yield was obtained.

The enantioselective protocol could be easily scaled up to the 2 mmol scale and afforded compound **3ba** in 89% yield with 92% ee (Scheme 2). It is noteworthy that by utilizing OH- and CH-based modifiable sites, axially chiral ArOBIN skeletons can be easily transformed into various synthetically useful intermediates, chiral ligands, and catalysts (see the

Scheme 2. Scalable Synthesis and Synthetic Elaborations



Supporting Information).[26] To further demonstrate the usage of this enantioselective cross-coupling, we conducted further derivations. For example, with potassium carbonate as the base, the methylation was accomplished by using methyl iodide in acetone and furnished compound **4** in quantitative yield with 91% ee. Moreover, the versatile intermediate **5** was obtained via a tosylation process under mild conditions in 96% yield without a decrease in ee.

On the basis of the obtained results and previous reports,[3d,6c] a plausible mechanism is depicted in Scheme 3A. The monoligand coordinated nickel complex **A** was formed as an active chiral catalyst from Ni(cod)$_2$ in the presence of ligand **L9** (1.2 equiv). According to the literature precedents, substrate **1a** had a slightly twisted structure with a molecular torsion angle, and **1aa** and **1ab** were in rapid equilibrium with each other.[30] Thus two possible nickelacycles **B** and **B′** could be formed via the oxidative addition.[31] Subsequent transmetalation with the aryl Grignard reagent occurred and generated the high-valent nickel species **C**. The desired products **1** were furnished via reductive elimination, which fulfilled a catalytic cycle. In the proposed mechanism, oxidative addition and transmetalation were the two possible steps to facilitate the enantioselectivity, and thus control experiments were conducted to determine the possibility of interconversion between nickelacycles **B** and **B′**. Compounds (*S*)-2′-hydroxy[1,1′-binaphthalen]-2-yl-trifluoromethanesulfonate (**6aa**) and (*R*)-2′-hydroxy[1,1′-binaphthalen]-2-yl-trifluoromethanesulfonate (**6ab**) were prepared and submitted to the catalytic system, and the corresponding products **3aa** and **3aa′** were formed in comparable yields via an enantioretentive process (Scheme 3B).[32] The observed independence of the output on the chiral nickel catalyst may suggest that the stereochemical outcome is determined at the oxidative addition step.

In summary, by employing a chiral NHC ligand, we demonstrated the first enantioselective cross-coupling reaction of inert aromatic C−O electrophiles. The chiral NHC ligand plays a pivotal role in this process by facilitating the inert C−O bond activation under mild conditions and controlling the enantioselectivity. This transformation was accomplished via

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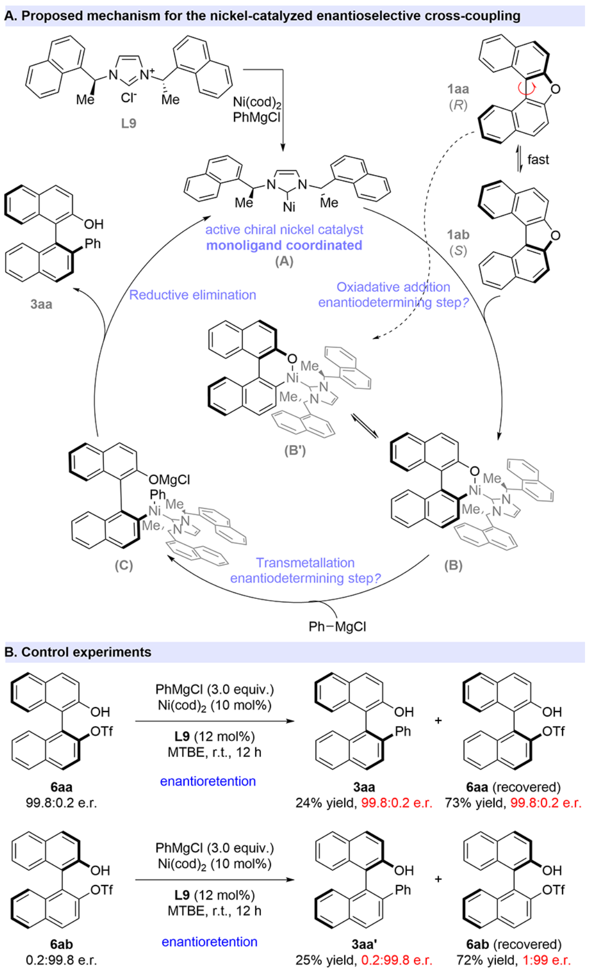
Scheme 3. Plausible Mechanism and Related Control Experiments



an enantioselective ring-opening process and offers a complementary entry to the construction of versatile axially chiral biaryl skeletons with good atom and step economy. The sequential derivations were elaborated by using CH- and OH-based modifiable sites. Efforts to reveal the detailed mechanism, to extend the substrate scope, and to improve the efficiency of this enantioselective cross-coupling are under way.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.1c09797.

Experimental procedures, characterization data of compounds, determination of the absolute configuration, NMR spectra, and HPLC traces (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We are grateful for financial support from Anhui Agricultural University. Prof. Mingyu Yang (Shanxi Normal University) is acknowledged for the HRMS analysis.

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